Weekend and Weekday Airbnb Prices Across Ten European Cities

STA302H1 Fall 2026 Group 199
Part 1 Research Proposal

Contributions

Norris (Yicheng) Wang was responsible for data verification, RQ exploration, and part of the literature exploration. Besides, he fitted the preliminary model, produced the summary tables, coefficient estimates and diagnostic figures, and conducted the initial model validation check.

Samario (Xiaoshuai) Liu was responsible for Data Description and Preliminary Results sections, including variable summaries, regression interpretation, assumption checks, and literature comparisons.

Suzy (Rui) Su wrote the Introduction, Ethics Discussion and Plan sections. Rui verified the three peer-reviewed sources against the original articles. Rui also prepared the project schedule and Gantt chart. In addition, Rui reviewed the other sections against the rubric and help to revise the interpretation diagnostic discussion.

1 Introduction

Airbnb prices can differ between weekday and weekend stays. Studying whether the price difference is similar for both options is significant for travellers comparing entire homes and private rooms. Our research question is: Across ten European cities, is the adjusted weekend–weekday difference in mean advertised prices the same for entire homes/apartments and private rooms? The response, realSum, is the total advertised price in euros for two guests staying two nights. The predictors of main focus are day type and room type, with adjustments for city, listing size, guest ratings, Superhost status, host portfolio size, and location.

Although raw group means may reflect differences in listing characteristics, multiple linear regression (MLR) better compares weekend and weekday prices within the same city while holding other measured characteristics fixed. We apply day type × room type interaction to estimate, in euros, how much the adjusted weekend–weekday price difference for entire homes/apartments differs from that for private rooms. The model allows weekend–weekday differences to vary by city while estimating one common interaction across cities. We test whether this interaction is zero and report a confidence interval to show the uncertainty in the estimate. Moreover, residual diagnostics will check whether the model assumptions are reasonable.

Earlier studies support our predictors and design. Gyódi and Nawaro (2021a), who collected our data, found that listing size, quality, and location were all associated with Airbnb prices across these ten cities, supporting our choice of variables. In Bristol, Voltes-Dorta and Sánchez-Medina (2020) found that models using property, host, and location characteristics fit entire-property prices better than private-room prices. This motivates checking whether the weekend–weekday difference also varies by room type. Casamatta et al. (2022) used host type × month interactions, and their OLS model found that listings from professional hosts had daily rates about 19% higher than those from non-professional hosts in August, with a much smaller gap outside peak months in Corsica. Their model allowed the price gap between host types to vary by month. We use a similar approach to test whether the weekend–weekday price difference varies by room type, which extends the original ten-city study.

The findings can help travellers understand how advertised accommodation costs differed by day type and room type in the sampled cities, and give hosts a reference when setting weekend prices. However, since the weekday and weekend samples are unpaired, the estimates only describe associations in the sampled market.

2 Data description

Gyódi and Nawaro (2021a) collected Airbnb offers to study prices and spatial dependence. We obtained their data from Zenodo Gyódi and Nawaro (2021b). Selenium searches captured prices for two guests staying two nights, four to six weeks before selected 2019 stays. Weekday stays ran Tuesday to Thursday and weekend stays Friday to Sunday.

After excluding 366 shared-room offers, we retained 51,341 records. The response, `realSum`, is the advertised total in euros, including reservation and cleaning fees. Its continuous scale allows us to model mean price differences. The mean (€280.85) exceeds the median (€212.23), consistent with right skew. Nearby listings and possible repeats raise concerns about independence. The files lack identifiers to check for repeats.

We use a day-type by room-type interaction to test whether the weekend–weekday price difference varies by room type. Prior research found that price relationships vary by room type Voltes-Dorta and Sánchez-Medina (2020) and that price gaps between host types vary by month Casamatta et al. (2022). City interactions allow day-type and room-type price differences to vary by city. City and day type come from filenames. Host portfolio combines the original `multi` and `biz` indicators.

Table 1: Numerical summaries: mean (SD); minimum to maximum. Categories: percentages. No missing values; ratings approach their ceilings. High prices and ten bedrooms warrant checking.

Variable	Summary
`realSum` (EUR; response)	280.85 (328.84); 34.78 to 18,545.45
`person_capacity` (maximum guests)	3.16 (1.30); 2 to 6
`bedrooms` (0 denotes studios)	1.16 (0.63); 0 to 10
`cleanliness_rating` (0–10)	9.39 (0.95); 2 to 10
`guest_satisfaction_overall` (0–100)	92.65 (8.93); 20 to 100
`dist` (city centre; km)	3.19 (2.39); 0.02 to 25.28
`metro_dist` (nearest metro; km)	0.68 (0.86); <0.01 to 14.27
`attr_index_norm` (within-city attraction accessibility; 0–100)	13.43 (9.81); 0.93 to 100
`city`	Amsterdam 4.03%; Athens 10.26%; Barcelona 5.49%; Berlin 4.70%; Budapest 7.81%; Lisbon 11.08%; London 19.37%; Paris 12.84%; Rome 17.56%; Vienna 6.86%
`day_type`	weekdays 49.32%; weekends 50.68%
`room_type`	Private room 36.41%; Entire home/apt 63.59%
`host_is_superhost`	Yes 25.69%; no 74.31%
`host_portfolio` (offers)	Single offer 35.96%; 2–4 29.20%; 5+ 34.84%

3 Ethics discussion

Based on the ethics module's trustworthiness criteria, we consider the dataset suitable. The data were collected, not simulated, since Gyódi and Nawaro (2021a) used automated Airbnb searches for two-person, two-night stays listing prices. Their peer-reviewed article describes the data source and collection process with a permanent DOI, and the Zenodo record documents the variables used in our analysis. The MD5 checksums of the 20 raw CSV files matched those in the archive, confirming that we downloaded the archived files intact.

However, the popularity of a dataset does not mean it is ethical to use. The researchers collected information from public Airbnb listings, so we do not know whether the hosts agreed to have their listings used for research. The dataset does not include host names or IDs, but the recorded coordinates could still help someone identify a property. To protect hosts' privacy, we report results for groups of listings and do not show individual locations.

The dataset was released under a CC BY 4.0 license, and we credit the authors and cite the Zenodo record. Therefore, we think this is an ethical way to use the data for the project.

4 Preliminary results

Raw-price OLS (12 predictors, n=51,341) estimates a €3.88 larger adjusted weekend–weekday mean price difference for entire homes than for private rooms within a city at fixed covariates (Table 2). The conventional 95% CI [−€7.33, €15.09] includes zero. Model diagnostics make inference provisional.

Positive bedroom (€81.00 per bedroom) and attraction (€6.58 per index point) slopes agree directionally with Gyódi and Nawaro (2021a). The centre-distance slope (−€0.36/km) is inconclusive, consistent with their less robust distance findings.

Table 2: OLS coefficients (conventional SEs), EUR. References: Amsterdam, weekdays, private rooms, non-Superhosts, single offers.

Term / city	Estimate (SE)	City × weekend (SE)	City × entire (SE)
Intercept	−12.34 (18.89)		
`day_type`: weekends	26.00 (12.92)		
`room_type`: entire home/apt	285.87 (12.93)		
`person_capacity`	27.80 (1.40)		
`bedrooms`	81.00 (2.51)		
`cleanliness_rating`	5.76 (1.90)		
`guest_satisfaction_overall`	0.93 (0.21)		
`host_is_superhost`: Superhost	4.96 (3.06)		
`dist`	−0.36 (0.94)		
`metro_dist`	−4.95 (1.88)		
`attr_index_norm`	6.58 (0.20)		
`host_portfolio`: 2–4 offers	8.47 (3.23)		
`host_portfolio`: 5+ offers	25.09 (3.28)		
`day_type` × `room_type`	**3.88 (5.72)**		
Athens	−216.28 (18.34)	−34.45 (14.91)	−302.27 (19.46)
Barcelona	−192.49 (13.23)	19.09 (16.67)	0.29 (18.67)
Berlin	−207.25 (14.25)	−10.78 (17.06)	−170.74 (17.43)
Budapest	−250.31 (17.98)	−7.86 (15.52)	−279.88 (19.35)
Lisbon	−180.26 (13.32)	−20.68 (14.62)	−238.46 (15.05)
London	−175.93 (12.03)	−28.83 (13.78)	−50.85 (13.92)
Paris	−88.02 (13.44)	−46.35 (14.40)	−182.06 (15.13)
Rome	−198.85 (12.07)	−23.31 (13.88)	−280.35 (14.10)
Vienna	−142.72 (15.53)	−28.09 (15.80)	−277.50 (17.15)

Additional conditions

- **Conditional mean response.** Binned means fall below the identity line in the midrange and rise above it at both ends (Figure 1a), suggesting $g(z)=z$ is inadequate.
- **Conditional mean predictor.** Attraction accessibility drops steeply then flattens with centre distance (Figure 1b), suggesting a nonlinear pairwise conditional mean.

Error assumptions

These departures limit what residual patterns can diagnose.

- **Linearity (mean-zero errors).** Binned residuals go from positive to negative and back (Figure 2, left), suggesting mean misspecification.
- **Constant error variance.** Residuals fan out at higher fitted prices (Figure 2, left), consistent with heteroskedasticity.
- **Normal errors.** The Q–Q upper tail lies above the reference line (Figure 2, right), suggesting a poor normal-error approximation.
- **Uncorrelated errors.** Neighbouring Paris cells share residual signs (Figure 3), consistent with spatial dependence or omitted location effects. Without stable listing or host IDs, independence remains unresolved.

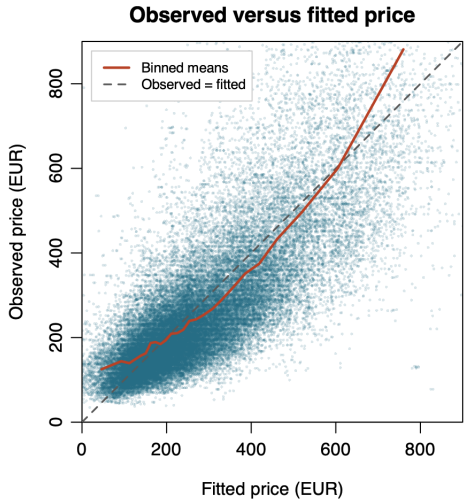


Figure 1a: Observed versus fitted prices (€0–€900). Binned means use all offers.

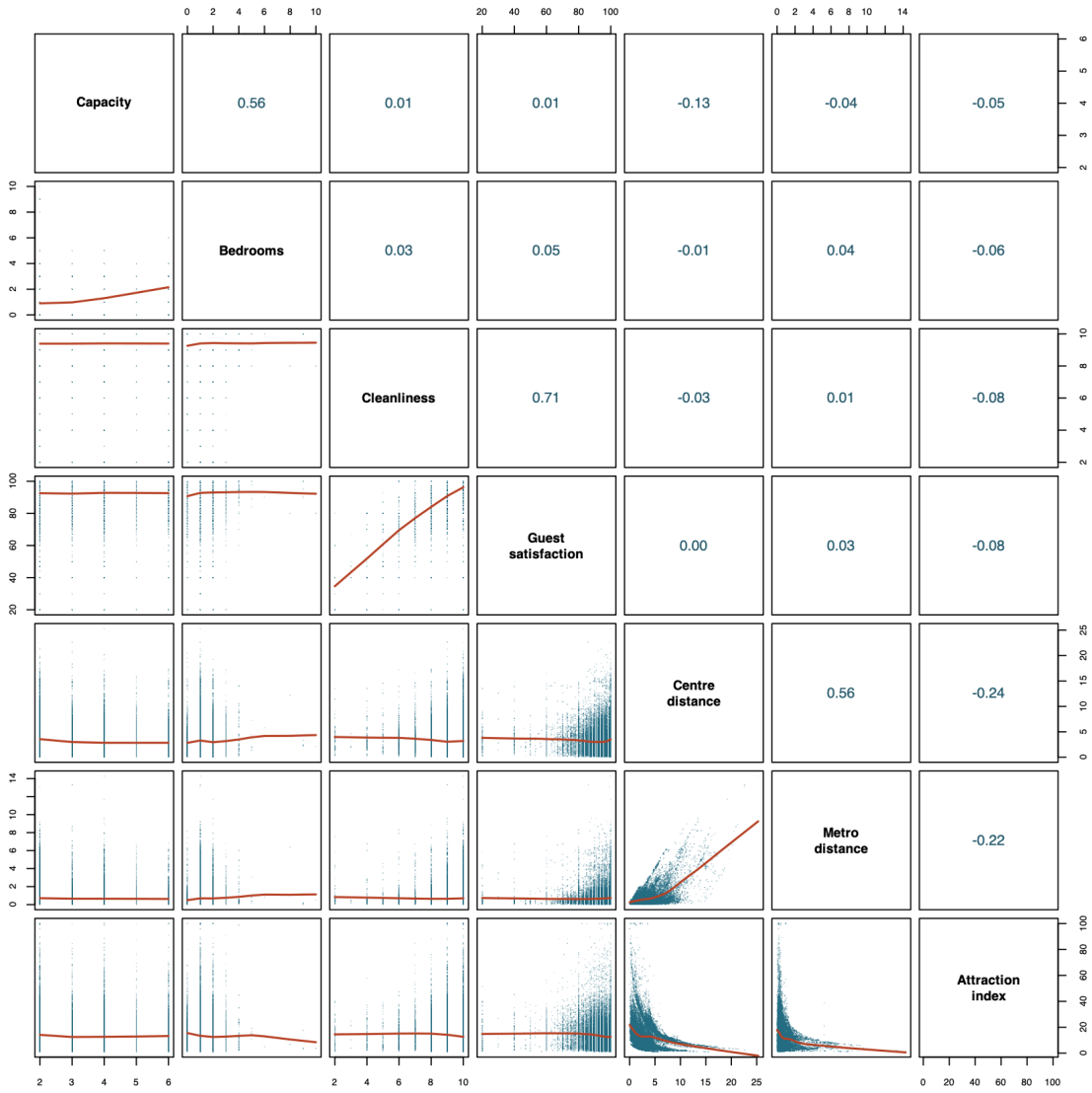


Figure 1b: Numerical predictor pairs and Pearson correlations.

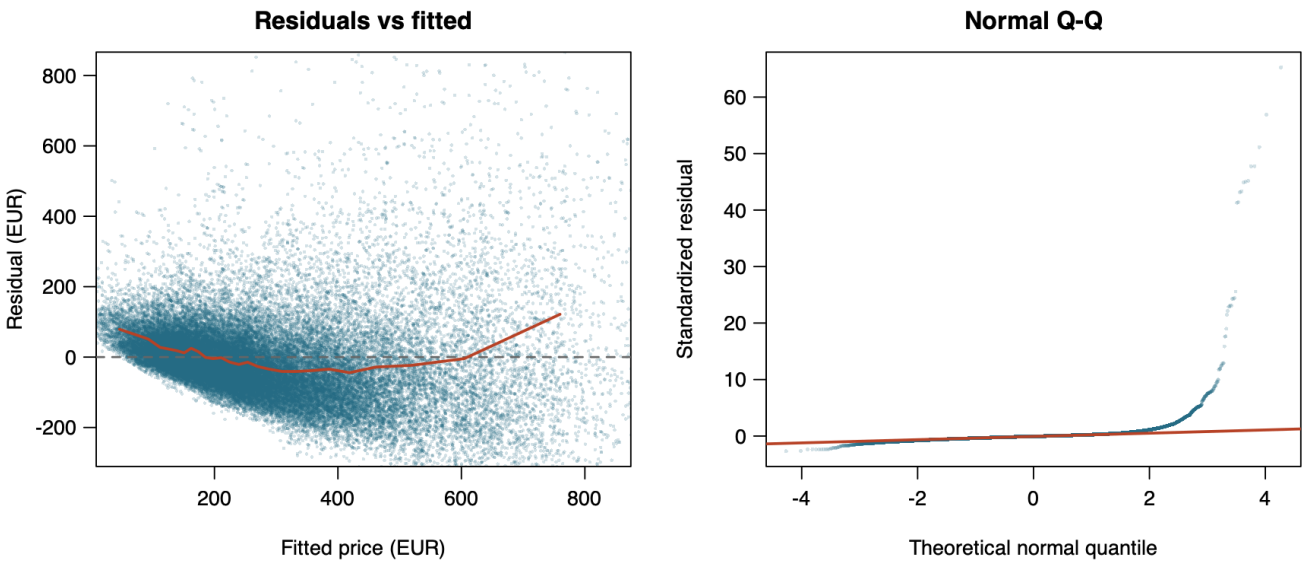


Figure 2: Residuals versus fitted (central 99% axis ranges) and full-range Normal Q–Q.

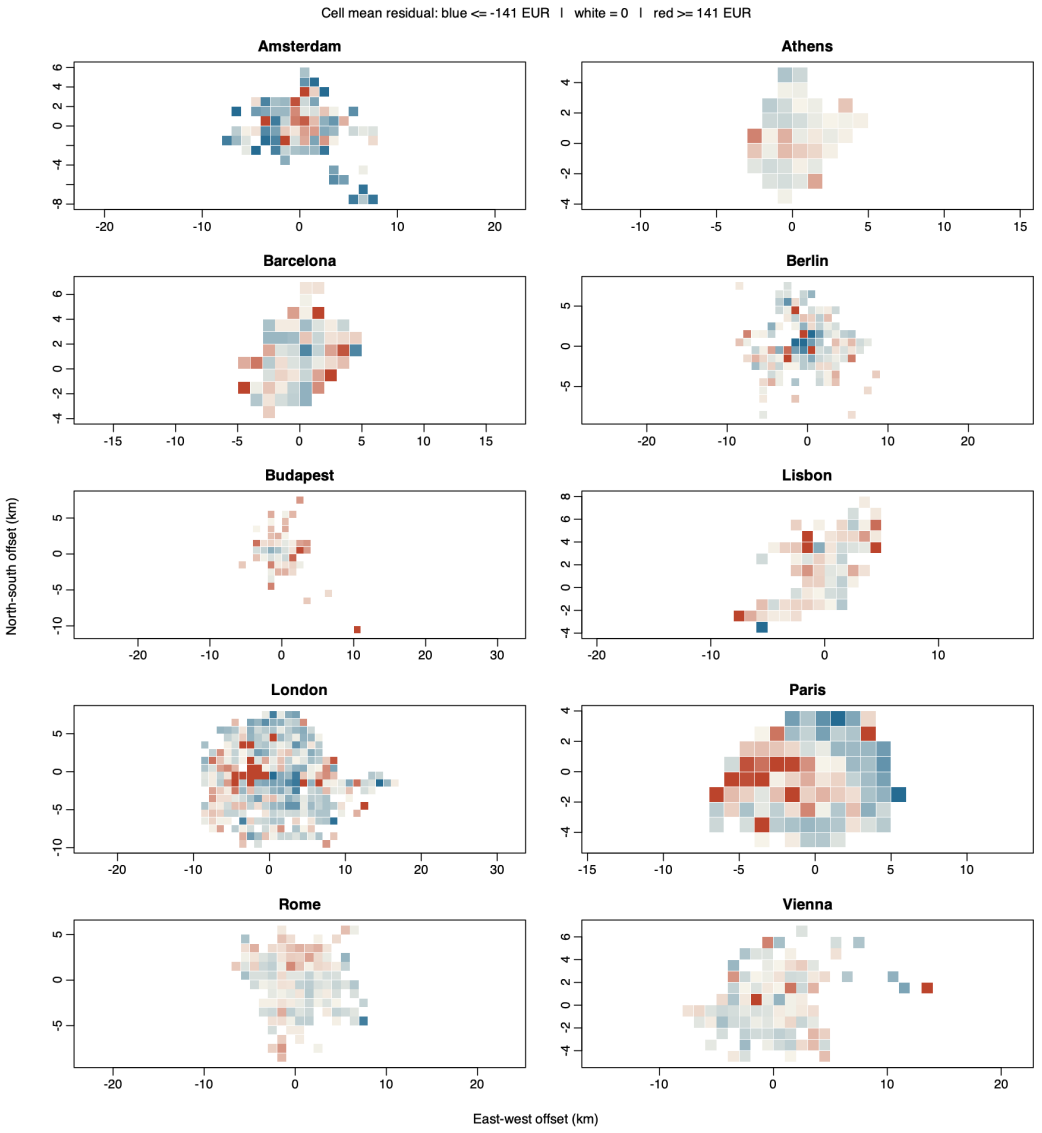


Figure 3: Mean residuals in approximately 1 km cells with ≥5 offers.

5 Plan

We will retain day type, room type, their interaction, and city terms to answer our research question. For other predictors, we will use coefficient t-tests and partial F-tests for groups of terms. Correlations between distance measures ($r = 0.56$) and ratings ($r = 0.71$) motivate comparing models retaining one or both measures. Literature and context will guide selection; we will compare adjusted R^2 on the same response scale and track the main interaction estimate.

All observed prices are positive, but 195 fitted prices are negative. Given the right-skewed response and unequal residual spread, we will consider Box–Cox transformations, likely log price, comparing residual patterns, spread, and Q–Q plots. With log price, we will compare weekend-to-weekday price ratios across room types. Predictor-specific curvature will motivate testing nonlinear terms. We will recheck all residual plots and both additional conditions after every model change.

Figure 4 gives responsibilities and milestones through December 2.

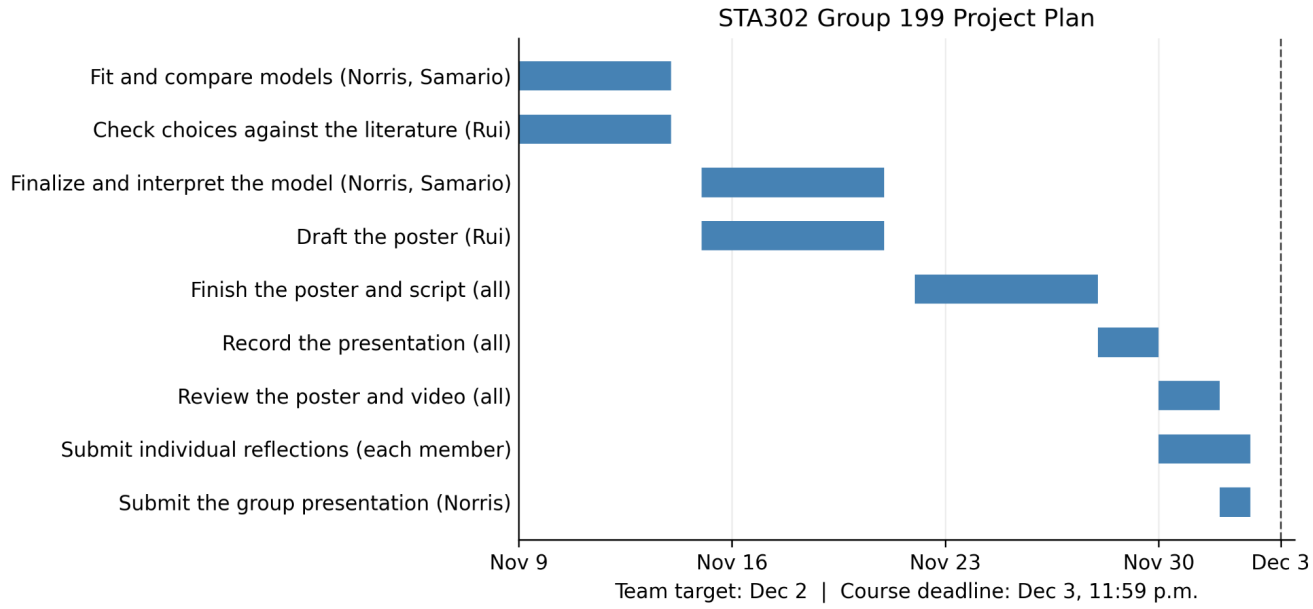


Figure 4: Team schedule and responsibilities.

Bibliography

Casamatta, G., Giannoni, S., Brunstein, D., & Jouve, J. (2022). Host type and pricing on Airbnb: Seasonality and perceived market power. *Tourism Management, 88*, 104433. https://doi.org/10.1016/j.tourman.2021.104433

Gyódi, K., & Nawaro, Ł. (2021a). Determinants of Airbnb prices in European cities: A spatial econometrics approach. *Tourism Management, 86*, 104319. https://doi.org/10.1016/j.tourman.2021.104319

Gyódi, K., & Nawaro, Ł. (2021b). *Determinants of Airbnb prices in European cities: A spatial econometrics approach (Supplementary Material)* (Version v2) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.4446043

Voltes-Dorta, A., & Sánchez-Medina, A. (2020). Drivers of Airbnb prices according to property/room type, season and location: A regression approach. *Journal of Hospitality and Tourism Management, 45*, 266–275. https://doi.org/10.1016/j.jhtm.2020.08.015